

Supplementary Material

Causal WeatherGraph: screen–confirm–replicate discovery of lagged wind–humidity–cloud dependencies in global atmospheric reanalysis

Xinchen Geng, University of Southern California

This document contains Supplementary Tables S1–S17 and a description of the machine-readable Supplementary Data. Tables S2–S6, S9 and S10 report the full-record screening stage (1979–2025 climatology, own-history OLS F -tests with BH within regime $\times$ target-variable families); they document the reproducibility of the screen and are not evidence of structural specificity, which is assessed by the confirmation and replication stages in the main text. Tables S2–S6, S9 and S10 (panels A and C–E) reproduce analyses of the original submission, which used the earlier, interpolated 2023–2025 continuation (Table S1, panel A); rerunning the full-record regime screen on the re-acquired record changes the significance of 670 of its 18,018 tests (3.7%) and the number significant from 14,274 to 14,308.

S1. Data provenance

Table S1: Same-date comparison of two CDS acquisition routes with the WeatherBench 2 conservative product on 152 timestamps (2022 January, April, July and October days 1–7; 2023 January days 1–10). Panel A: fields requested on the CDS 5.625° grid, which interpolates, as used for 2023–2025 in the original submission. Panel B: native 0.25° fields regridded with the WeatherBench 2 first-order conservative scheme, as used for 2023-01-11 to 2025-12-31 in this revision. Grid values are compared in physical units and as discovery-standardized anomalies; regional values are the 66 regional means of standardized anomalies that enter the regressions.

Variable	Grid, physical units		Grid Correlation	Regional, standardized		
	Mean CDS–WB2	RMSE		Mean CDS–WB2	RMSE	Correlation
<i>A. CDS 5.625° interpolation (original submission)</i>						
Temperature T	-0.057 K	0.982	0.949	-0.012	0.080	0.991
Specific humidity q	-0.010 g kg ⁻¹	0.841	0.874	-0.026	0.141	0.963
Wind speed W	+0.685 m s ⁻¹	2.025	0.869	+0.198	0.243	0.954
Zonal wind u	+0.003 m s ⁻¹	2.012	0.909	-0.004	0.125	0.967
Meridional wind v	+0.010 m s ⁻¹	1.649	0.914	+0.008	0.107	0.979
Total cloud cover C	+0.119 %	18.382	0.802	+0.010	0.199	0.907
<i>B. Native 0.25° fields, conservative regridding (this revision)</i>						
Temperature T	+4.1 × 10 ⁻⁶ K	5.6 × 10 ⁻⁵	1.000	+3.7 × 10 ⁻⁶	1.3 × 10 ⁻⁵	1.000
Specific humidity q	+2.7 × 10 ⁻⁸ g kg ⁻¹	4.9 × 10 ⁻⁶	1.000	-2.2 × 10 ⁻⁷	1.4 × 10 ⁻⁵	1.000
Wind speed W	+4.1 × 10 ⁻⁷ m s ⁻¹	1.9 × 10 ⁻⁵	1.000	+1.2 × 10 ⁻⁷	1.5 × 10 ⁻⁶	1.000
Zonal wind u	-2.2 × 10 ⁻⁷ m s ⁻¹	1.8 × 10 ⁻⁵	1.000	-4.8 × 10 ⁻⁸	1.1 × 10 ⁻⁶	1.000
Meridional wind v	+5.7 × 10 ⁻⁸ m s ⁻¹	1.9 × 10 ⁻⁵	1.000	+3.7 × 10 ⁻⁸	1.6 × 10 ⁻⁶	1.000
Total cloud cover C	+2.5 × 10 ⁻⁵ %	8.5 × 10 ⁻⁵	1.000	+1.3 × 10 ⁻⁶	1.9 × 10 ⁻⁶	1.000

S2. Robustness of the screening stage

Table S2: Historical-period screening (own-history OLS, full-record climatology).

Period	Time range	Significant ratio	WH edges	HC edges	WC edges	WHC chains	Overlap with full graph
P1	1979-1989	0.734	428	407	421	2,700	0.910
P2	1990-1999	0.714	428	391	411	2,647	0.901
P3	2000-2009	0.724	424	390	427	2,650	0.910
P4	2010-2018	0.715	406	388	418	2,531	0.906
P5	2019-2025	0.678	390	365	386	2,261	0.883

Table S3: Candidate-neighbourhood expansion.

k	Candidate edges	Tested edges	Significant ratio	Unique significant edges	WH edges	HC edges	WHC chains	Recovered k=2 edges
2	858	18,018	0.792	857	3,260	3,079	22,202	857
4	1,386	29,106	0.746	1,379	4,956	4,926	54,351	857
6	1,914	40,194	0.697	1,892	6,400	6,655	95,871	857
8	2,442	51,282	0.672	2,403	7,804	8,376	148,573	857

Table S4: Preprocessing variants.

Setting	Significant ratio	WH edges	HC edges	WC edges	WHC chains	WHC chain overlap
P1_monthly	0.792	3,260	3,079	3,260	22,202	1.000
P2_monthly_diff	0.874	3,535	3,630	3,444	27,922	0.856
P3_diff_only	0.878	3,546	3,676	3,442	28,468	0.858
P4_rolling	0.852	3,517	3,427	3,415	26,472	0.854
P5_monthly_regionstd	0.784	3,180	3,141	3,168	22,131	0.883

Table S5: Regionalization variants.

Setting	Method	Actual regions	Significant ratio	WH edges	HC edges	WC edges	WHC chains	Midlatitude strongest
R32	latlon_bins	32	0.785	1,491	1,498	1,573	10,245	no
R64	latlon_bins	66	0.792	3,260	3,079	3,260	22,202	yes
R128	latlon_bins	128	0.822	6,460	6,488	6,358	47,309	yes
K64	kmeans_sphere	64	0.791	3,035	3,161	3,159	21,766	yes
K128	kmeans_sphere	128	0.804	6,075	6,474	6,359	44,219	yes

S3. Regime screening, external states and resampling

Table S6: Descriptive regime screening: chain statistics, graph distances and matched-sample sensitivity. These quantities compare separately selected graphs and are not tests of state dependence.

A. Lag-resolved wind–humidity–cloud chains

Regime	WHC chains	Intermediate humidity regions	Mean total lag	Mean chain score
all	3,746	66	3.983	0.468
DJF	3,215	66	3.976	0.578
JJA	2,998	66	3.963	0.553
high_humidity	2,442	62	3.939	0.551
normal_humidity	3,522	65	3.988	0.486
high_cloud	2,612	65	3.962	0.499
normal_cloud	3,667	66	3.972	0.468

B. Jaccard distances between regime graphs

Comparison	Jaccard distance	Union edges	Intersection edges
DJF vs JJA	0.107	851	760
high_humidity vs normal_humidity	0.106	840	751
high_cloud vs normal_cloud	0.087	841	768
all vs DJF	0.075	856	792
all vs JJA	0.065	851	796

C. Matched-sample sensitivity

Comparison	High samples	Normal samples	Original Jaccard	Matched Jaccard mean	Matched Jaccard SD	Mean edge-count difference
high humidity vs normal humidity	13,734	41,200	0.106	0.119	0.006	-0.75
high cloud vs normal cloud	13,734	54,934	0.087	0.111	0.007	15.50

Table S7: Niño-3.4 state contrasts in 2019–2025 (36 discovery-selected region pairs, lags 1–3, HAC bandwidth 64). Entries are medians over pairs and lags of the signed and absolute slope differences, without and with the six physical controls taken at $t - 4$ to $t - 6$.

Contrast	Edge type	Own history		Own history + physical controls		
		Median $\Delta\beta$	Median $ \Delta\beta $	Median $\Delta\beta$	Median $ \Delta\beta $	Median SE
High – middle	$W \rightarrow H$	+0.0038	0.0078	+0.0041	0.0068	0.0080
High – middle	$H \rightarrow C$	-0.0001	0.0101	-0.0008	0.0147	0.0134
High – middle	$W \rightarrow C$	+0.0067	0.0092	+0.0054	0.0073	0.0121
Sign changes after adding physical controls: 30 of 108 contrasts						
Low – middle	$W \rightarrow H$	-0.0002	0.0089	-0.0018	0.0097	0.0070
Low – middle	$H \rightarrow C$	-0.0006	0.0061	+0.0081	0.0141	0.0135
Low – middle	$W \rightarrow C$	-0.0006	0.0144	-0.0002	0.0078	0.0107
Sign changes after adding physical controls: 24 of 108 contrasts						

The original stability analysis drew 20 block-bootstrap replicates from the list of valid times of each regime. For non-contiguous regimes this concatenated separated dates and rebuilt lags across the joins; its stability counts are therefore not reproduced here and are superseded by the calendar-preserving resampling in Table S8 and, inferentially, by the held-out replication of the main text.

Table S8: Calendar-preserving block bootstrap for twelve fixed edge-regime models (200 replicates per block length). Lagged design rows are formed on the original calendar before blocks are resampled, so no lag spans a block boundary. Intervals are percentile 95% intervals; the last column is the share of positive replicates at block length 128.

Regime	Regions	Edge	Lag	Median β	Block 64	Block 128	Block 256	Positive
all	0→0	$W \rightarrow H$	1	+0.0048	[+0.0032, +0.0066]	[+0.0031, +0.0063]	[+0.0031, +0.0063]	1.000
all	0→0	$H \rightarrow C$	1	+0.0256	[+0.0220, +0.0290]	[+0.0219, +0.0283]	[+0.0221, +0.0294]	1.000
all	22→22	$W \rightarrow H$	1	+0.0001	[-0.0012, +0.0015]	[-0.0009, +0.0016]	[-0.0014, +0.0017]	0.595
all	22→22	$H \rightarrow C$	1	+0.0347	[+0.0312, +0.0383]	[+0.0309, +0.0389]	[+0.0300, +0.0401]	1.000
all	44→44	$W \rightarrow H$	1	-0.0069	[-0.0088, -0.0053]	[-0.0084, -0.0050]	[-0.0088, -0.0052]	0.000
all	44→44	$H \rightarrow C$	1	-0.0172	[-0.0206, -0.0135]	[-0.0203, -0.0136]	[-0.0208, -0.0125]	0.000
high humidity	0→0	$W \rightarrow H$	1	+0.0008	[-0.0027, +0.0042]	[-0.0025, +0.0042]	[-0.0025, +0.0044]	0.685
high humidity	0→0	$H \rightarrow C$	1	+0.0260	[+0.0190, +0.0315]	[+0.0198, +0.0341]	[+0.0193, +0.0340]	1.000
high humidity	22→22	$W \rightarrow H$	1	+0.0023	[-0.0003, +0.0053]	[-0.0003, +0.0054]	[-0.0005, +0.0054]	0.960
high humidity	22→22	$H \rightarrow C$	1	+0.0231	[+0.0153, +0.0317]	[+0.0138, +0.0317]	[+0.0132, +0.0333]	1.000
high humidity	44→44	$W \rightarrow H$	1	-0.0056	[-0.0102, -0.0014]	[-0.0098, -0.0013]	[-0.0105, -0.0012]	0.000
high humidity	44→44	$H \rightarrow C$	1	-0.0138	[-0.0205, -0.0060]	[-0.0207, -0.0067]	[-0.0213, -0.0066]	0.000

S4. Earlier screening controls

Table S9: Graph null models, enrichment and directional falsification for the full-record screen.

A. Random candidate controls (single draws)

Edge set	Tested edges	Significant edges	Significant ratio	Mean absolute coefficient	Mean effect score
physical_candidates	18,018	14,274	0.792	0.0228	0.2446
random_edges	18,018	9,235	0.513	0.0186	0.1814
distance_matched_random	18,018	13,392	0.743	0.0208	0.2200
variable_preserved_random	18,018	8,244	0.458	0.0189	0.1813

B. Enrichment under variable-preserved randomization

Edge type	True significant ratio	Random-region significant ratio	Enrichment
Wind → Humidity	0.784	0.444	1.77
Humidity → Cloud Cover	0.741	0.440	1.68
Wind → Cloud Cover	0.784	0.406	1.93

C. Directionality and pseudo-time controls

Test	Edge type	Tested edges	Significant edges	Significant ratio	Mean effect score
forward	Wind → Humidity	4,158	3,247	0.781	0.207
reverse	Humidity → Wind	4,158	2,655	0.639	0.123
time_shuffled	Wind → Humidity	4,158	0	0.000	0.000
circular_shift	Wind → Humidity	4,158	1,498	0.360	0.115
forward	Humidity → Cloud Cover	4,158	3,076	0.740	0.284
reverse	Cloud → Humidity	4,158	3,332	0.801	0.189
time_shuffled	Humidity → Cloud Cover	4,158	1	0.000	0.063
circular_shift	Humidity → Cloud Cover	4,158	1,585	0.381	0.120
forward	Wind → Cloud Cover	4,158	3,258	0.784	0.242
time_shuffled	Wind → Cloud Cover	4,158	0	0.000	0.000

Table S10: Latitude bands, long lags, humidity adjustment, effect sizes and PCMCi agreement for the full-record screen.

A. Latitude-band chain statistics

Latitude band	WHC chains	Stable WHC chains	Mean total lag	Mean chain score
tropical	6,017	1,201	3.985	0.451
midlatitude	9,612	3,628	3.983	0.642
polar	6,573	888	3.941	0.375

B. Long-lag screen with maximum lag 12 (all-time regime, own-history OLS, BH within target-variable families; recomputed with the revision code on the re-acquired record)

Edge type	Significant	Peak lag	Mean lag	Median lag	Lag 1	Lags 2–3	Lags 4–8	Lags 9–12
Wind → Humidity	1,683	4	6.11	6.0	0.092	0.190	0.434	0.284
Humidity → Cloud Cover	1,744	4	6.20	6.0	0.092	0.179	0.432	0.296
Wind → Cloud Cover	1,829	2	6.33	6.0	0.089	0.182	0.415	0.315
Humidity → Humidity	1,317	3	6.33	6.0	0.087	0.179	0.424	0.310

C. Coefficient change of wind→cloud after adding target-region humidity

Regime	WC edges	Mean direct beta	Mean beta after controlling Humidity	Mean reduction	Reduction > 30%	Reduction > 50%
all	520	-0.00156	-0.00143	0.0369	0.206	0.117
DJF	436	-0.00293	-0.00191	0.0729	0.211	0.108
JJA	482	-0.00150	-0.00218	0.0330	0.199	0.100
high_humidity	429	-0.00157	-0.00140	0.1045	0.182	0.091
normal_humidity	492	-0.00163	-0.00126	0.0873	0.215	0.118
high_cloud	391	-0.00098	-0.00069	0.1090	0.192	0.079
normal_cloud	510	-0.00155	-0.00137	0.0358	0.198	0.102

D. Target-region multivariable adjustment and effect sizes

Edge type	Principal significant	Adjusted significant	Original retained	Recovery	Median $ \beta $	Median partial R^2	Top-edge block CI exclusion
Wind → Humidity	503	490	439	0.873	0.0138	0.00117	10/10
Humidity → Cloud Cover	488	483	407	0.834	0.0207	0.00090	10/10
Wind → Cloud Cover	520	487	428	0.823	0.0176	0.00084	10/10

E. PCMCi agreement for 60 selected strong edges, 2019–2025

Edge type	Granger top edges	PCMCi confirmed	Confirmed ratio	Mean absolute partial correlation	Status
Wind → Humidity	20	18	0.900	0.0867	confirmed
Humidity → Cloud Cover	20	16	0.800	0.0426	confirmed
Wind → Cloud Cover	20	16	0.800	0.0815	confirmed

S5. Additional results for the revised analyses

Table S11: Three-stage results with month $\times$ UTC-hour anomalies (diurnal cycle removed), with the monthly-climatology results for comparison.

Edge type	Month $\times$ UTC-hour climatology					Monthly climatology	
	Screened	Confirmed	Replicated	Rate	Control rate	Replicated	Rate
$W \rightarrow H$	455	291	117	0.402	0.083	117	0.360
$H \rightarrow C$	460	291	122	0.419	0.122	133	0.425
$W \rightarrow C$	469	288	96	0.333	0.052	113	0.363
$H \rightarrow H$	338	270	147	0.544	0.167	153	0.560
$T \rightarrow H$	178	164	133	0.811	0.206	137	0.801
$T \rightarrow C$	164	153	106	0.693	0.244	107	0.682
<i>All</i>	2,064	1,457	721	0.495	0.105	760	0.490

Table S12: Time-aligned path models $W_i(t - \ell_1 - \ell_2) \rightarrow H_h(t - \ell_2) \rightarrow C_j(t)$ for 60 discovery-selected paths (20 per stratum), evaluated on 2019–2025 with frozen coefficients. Entries are the number of paths with positive held-out gain and, in parentheses, the median held-out ΔR^2 ($\times 10^{-4}$).

Stage	Baseline histories			+ six physical controls before the path source		
	Strong	Ordinary	Unselected	Strong	Ordinary	Unselected
$W_i \rightarrow H_h$	20 (5.18)	18 (1.20)	14 (0.13)	19 (1.00)	16 (0.26)	15 (0.06)
$H_h \rightarrow C_j \mid W_i$	19 (16.40)	14 (0.98)	15 (0.48)	17 (5.32)	13 (0.42)	17 (2.30)
$W_i \rightarrow C_j \mid H_h$	14 (1.67)	16 (1.45)	17 (0.37)	14 (0.61)	17 (0.59)	14 (0.27)
$W_i \rightarrow C_j$ total	16 (1.47)	17 (1.43)	17 (0.49)	16 (0.79)	16 (0.54)	15 (0.32)

Table S13: Known-structure benchmark details (six variables, 4,096 steps, 50 replicates per scenario; 90 scored edge-lag hypotheses). FDP is the mean false-discovery proportion against the generating direct graph.

Scenario	Method	Precision	Recall	F1	SHD	FDP
Weak AR	Own-history OLS	0.493	1.000	0.654	6.68	0.507
	Own-history HAC	0.475	1.000	0.639	7.10	0.525
	All-history OLS	0.975	1.000	0.986	0.18	0.025
	All-history HAC	0.956	1.000	0.976	0.32	0.044
	PCMCI	0.959	1.000	0.978	0.30	0.041
Strong AR	Own-history OLS	0.086	1.000	0.158	64.58	0.914
	Own-history HAC	0.085	1.000	0.157	65.02	0.915
	All-history OLS	0.945	1.000	0.970	0.40	0.055
	All-history HAC	0.940	1.000	0.966	0.46	0.060
	PCMCI	0.950	1.000	0.973	0.36	0.050
Observed driver	Own-history OLS	0.077	0.990	0.143	71.80	0.923
	Own-history HAC	0.077	0.990	0.142	72.06	0.923
	All-history OLS	0.954	1.000	0.974	0.38	0.046
	All-history HAC	0.938	1.000	0.965	0.50	0.062
	PCMCI	0.924	1.000	0.956	0.64	0.076
Hidden driver	Own-history OLS	0.077	0.990	0.143	71.80	0.923
	Own-history HAC	0.077	0.990	0.142	72.06	0.923
	All-history OLS	0.149	1.000	0.259	34.66	0.851
	All-history HAC	0.148	1.000	0.258	34.98	0.852
	PCMCI	0.145	1.000	0.252	35.90	0.855
Regime switching	Own-history OLS	0.699	1.000	0.816	2.92	0.301
	Own-history HAC	0.680	1.000	0.801	3.24	0.320
	All-history OLS	0.955	1.000	0.976	0.32	0.045
	All-history HAC	0.921	1.000	0.957	0.58	0.079
	PCMCI	0.957	1.000	0.976	0.32	0.043

Table S14: Degree- and distance-preserving topology randomization. On the original symmetric support, fixing every node’s type-specific in- and out-degree and the distance histogram admits only three (full record) or two (late versus early) feasible graphs, and the number of two-step chains equals $\sum_h d_{WH}^{\text{in}}(h) d_{HC}^{\text{out}}(h)$ for every feasible graph (651 and 627), so it cannot measure enrichment. Markov chains on an expanded support change few edges and did not mix adequately.

Case	Feasible graphs, original support	Distinct graphs, expanded support	Mean edges changed (%)	Productive acceptance (%)
Full record, symmetric support	3	778	4.30	0.016
2002–2025 versus 1979–2001 overlap	2	781	4.13	0.017

Table S15: Agreement between ERA5 and CERES SYN1deg Edition 4.2 cloud anomalies (monthly climatology; regional column under month $\times$ UTC-hour climatology also shown). Correlations are medians over grid cells or over the 66 regions; the bias is the area-weighted mean of CERES minus ERA5 total cloud cover in percentage points.

Window	Times	Grid-cell correlation				Regional correlation		Bias	
		All	Tropics	Midlat.	Polar	Monthly [range]	Diurnal		
2017–2025		13,147	0.759	0.798	0.814	0.567	0.763 [0.39, 0.91]	0.768	+3.83
2017-01-01 to 2023-01-10 (WB2 route)		8,803	0.758	0.797	0.808	0.564	0.763 [0.37, 0.91]	0.774	+4.01
2023-01-11 to 2025-12-31 (CDS route)		4,344	0.764	0.799	0.827	0.580	0.778 [0.38, 0.91]	0.783	+3.47

S6. Wind–humidity eddy convergence

Table S16: Pre-specified wind–humidity eddy convergence (WHEC) test: region-normalized frozen 1979–2018 block gains, that is, the percentage reduction of the held-out mean squared error when lags 1–3 of the test series are added to the dense VAR(3) with local u , v and linear convergence, averaged over the 22 regions of each latitude band, with HAC (bandwidth 64) z -statistics. Endpoint E1 is the midlatitude cloud row and E2 the midlatitude-minus-tropical cloud row of the local family; the remote and placebo families are the controls.

Family	Target	Band statistic	2019-01-01 to 2023-01-10		2023-01-11 to 2025-12-31	
			Gain (%)	z	Gain (%)	z
Local	Cloud	Tropical	0.149	6.1	0.144	5.7
		Midlatitude	0.456	10.6	0.359	6.8
		Polar	0.501	10.8	0.623	12.0
		Mid. – trop.	0.307	6.6	0.215	3.8
		Mid. – polar	−0.046	−0.8	−0.264	−3.3
		Polar – trop.	0.352	6.6	0.479	8.2
Local	Humidity	Tropical	1.502	18.1	1.226	16.4
		Midlatitude	5.070	32.9	4.661	23.5
		Polar	7.332	11.9	7.484	11.5
		Mid. – trop.	3.568	22.4	3.435	15.8
		Mid. – polar	−2.262	−3.4	−2.822	−4.1
		Polar – trop.	5.830	9.2	6.258	9.3
Remote	Cloud	Tropical	−0.005	−0.7	−0.002	−0.3
		Midlatitude	−0.001	−0.2	−0.011	−1.8
		Polar	0.002	0.4	−0.005	−0.8
		Mid. – trop.	0.003	0.4	−0.009	−0.9
		Mid. – polar	−0.003	−0.4	−0.007	−0.8
		Polar – trop.	0.007	0.8	−0.002	−0.2
Remote	Humidity	Tropical	0.006	1.0	0.013	1.8
		Midlatitude	−0.008	−1.2	−0.002	−0.3
		Polar	−0.005	−0.7	0.003	0.4
		Mid. – trop.	−0.013	−1.6	−0.015	−1.5
		Mid. – polar	−0.003	−0.3	−0.005	−0.5
		Polar – trop.	−0.011	−1.2	−0.011	−1.0
Placebo	Cloud	Tropical	−0.001	−0.2	0.000	0.0
		Midlatitude	−0.002	−0.4	−0.016	−2.2
		Polar	−0.007	−1.5	−0.016	−2.5
		Mid. – trop.	−0.001	−0.1	−0.016	−1.3
		Mid. – polar	0.005	0.6	0.000	0.0
		Polar – trop.	−0.006	−0.7	−0.016	−1.3
Placebo	Humidity	Tropical	0.011	1.2	0.023	2.3
		Midlatitude	−0.004	−0.7	−0.005	−0.8
		Polar	−0.001	−0.1	0.001	0.2
		Mid. – trop.	−0.015	−1.3	−0.028	−2.4
		Mid. – polar	−0.003	−0.3	−0.006	−0.7
		Polar – trop.	−0.012	−1.1	−0.022	−1.8

Table S17: Local WHEC by latitude band (66 lag-specific hypotheses per band and target). Confirmed: hypotheses passing Stages 1 and 2 (positive coefficients in parentheses); Rep./pred.: confirmed hypotheses that replicate with the same sign and that give a significant frozen predictive gain; Positive gain: regions of 22 with a positive frozen block gain (median gain in %); Wald: regions whose block Wald test in the held-out refit has BH $q < 0.05$ over 66 regions; CERES: hypotheses with the discovery sign and $q < 0.05$ in refits with CERES cloud as predictor and outcome.

Target	Band	Screened	Confirmed	Rep./pred.		Positive gain		Wald		CERES	
				2019–23	2023–25	2019–23	2023–25	2019–23	2023–25	2017–23	2023–25
Cloud	Tropical	44	30 (18)	15 / 7	11 / 2	19 (0.108)	17 (0.077)	10	6	18	19
	Midlatitude	59	50 (31)	28 / 9	20 / 10	22 (0.414)	20 (0.331)	18	15	24	6
	Polar	65	59 (38)	43 / 23	35 / 21	22 (0.424)	21 (0.590)	21	18	22	16
Humidity	Tropical	65	56 (32)	36 / 18	34 / 17	22 (1.266)	21 (1.312)	22	22	–	–
	Midlatitude	63	61 (37)	51 / 42	53 / 40	22 (5.411)	22 (4.129)	22	22	–	–
	Polar	61	61 (40)	52 / 43	54 / 45	22 (7.106)	22 (7.407)	22	22	–	–

S7. Supplementary Data

The machine-readable results are provided as comma-separated files.

- **S1_three_stage_all_candidates.csv**: all 2,574 edge-lag hypotheses with source and target regions and variables, lag, edge type, great-circle distance, Stage 1 and Stage 2 coefficients, p - and q -values and partial R^2 , held-out coefficients, directional replication q -values, frozen predictive gains with HAC standard errors for each held-out segment, and the stage indicators.
- **S2_symmetric_full_var_<period>.csv**: dense VAR(3) tests of the 3,816 symmetric edge-lag hypotheses in each period, with coefficients, HAC standard errors, p - and q -values and partial R^2 .
- **S3_regime_screening_ols_hac.csv**: the 18,018 full-record edge-lag-regime screening tests with OLS and HAC (bandwidths 32, 64, 128) inference, within-family and global BH q -values, effect sizes and residual autocorrelations.
- **S4_ceres_substitution_tests.csv**: own-history and dense VAR(3) tests of the 2,544 symmetric cloud hypotheses with ERA5 and with CERES cloud, by segment.
- **S5a_wb2_cds_route_variables.csv** and **S5b_wb2_cds_route_regions.csv**: variable-level and region-level differences between the WeatherBench 2 product and fields requested on the coarse CDS grid on 152 identical timestamps; **S5c_wb2_native_conservative_route.csv**: the same comparison for native fields regridded conservatively, the route used for 2023–2025 in this revision.
- **S6a_whec_design.json**: the design of the wind–humidity eddy convergence test, frozen before any of its results were computed (SHA-256 0507157c5a8ec0b2...).
- **S6b_whec_tests_per_lag.csv**: the 1,188 lag-specific WHEC, remote and placebo hypotheses with Stage 1–3 statistics; **S6c_whec_block_tests.csv**: block Wald tests and frozen block gains by family, target, region and period; **S6d_whec_band_endpoints.csv**: band means and differences of the normalized gains with HAC standard errors; **S6e_whec_ceres_check.csv**: refits with ERA5 and with CERES cloud on 2017–2023 and 2023–2025.

Region identifiers refer to the 66 regions of the 6×11 latitude–longitude partition, numbered by latitude band from south to north and by longitude bin from -180° eastward; their centroids are listed in **S0_regions.csv**.